

Assignment 9

Hilal Selen Bütün

2025-12-16

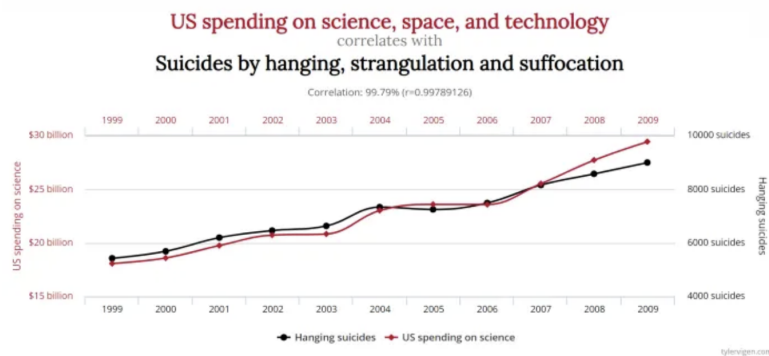


Figure 1: Analyzed Spurious Correlation Visualization

The main problem with this visualization is that it creates a misleading perception of causality for the viewer. The chart presents U.S. spending on science, space, and technology alongside the number of suicides by hanging and drowning on the same time axis. This presentation gives the impression that there is a meaningful relationship or a causal connection between the two variables. However, this relationship is purely a spurious correlation, and there is no causal link between the variables. Therefore, the visualization has the potential to mislead those who interpret it.

The primary technical issue in the chart is the arbitrary scaling of the dual axes. The designer of the visualization has deliberately adjusted the starting and ending values of the left and right Y-axes in order to make the lines representing two completely unrelated variables—expenditures measured in billions of dollars and suicide counts measured in number of individuals—visually overlap. As a result, a coincidental alignment is perceived by the viewer as a strong causal relationship.

The purpose of data visualization should be to reveal meaningful relationships. However, this chart emphasizes a meaningless and random overlap, making it a negative example from a data ethics perspective.

Solution 1: Showing Relative Change Instead of Absolute Values

In this visualization, the chart uses a dual-axis structure to show that science spending increased from 15 billion to 30 billion dollars, while the number of suicides rose from 4,000 to 10,000. The different units and scales of these two variables make it possible for the lines to be intentionally aligned.

To address this issue, both datasets can be indexed and standardized. For example, if the year 1999 is set to 100 for both variables, the Y-axis represents percentage change. In this case, the slopes of the lines diverge, allowing the viewer to clearly see that science spending increased at a faster rate than suicide counts. This removes the illusion of a “perfect overlap.”

Because science expenditures and suicide counts are measured in different units, using absolute values in line charts is inappropriate when comparing them. Instead, using relative changes helps prevent axis manipulation and allows for a more honest interpretation.

Solution 2: Preventing Contextual Misinterpretation

The fundamental goal of data visualization is to provide meaningful insights to the viewer. Presenting variables that have no causal or contextual relationship within the same chart leads to a false perception of causality.

For this reason, science spending should be visualized alongside economic and technological indicators, while suicide data should be presented with socio-economic and societal factors in separate charts. Avoiding the combination of unrelated variables in a single visualization results in a more accurate and ethical presentation of data.

Additionally, if this chart is intended solely as an example of a “spurious correlation,” it should be clearly stated in the title, captions, and explanatory text that the relationship is not causal.

Source: <https://www.tylervigen.com/spurious-correlations>